Generative Latent Space Modeling: Improvements and Interpretability on Fashion-MNIST

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Final Project Report

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Abstract

This report outlines the development and optimization of a generative model for the Fashion-MNIST dataset. We implement a Variational Autoencoder (VAE) from scratch, utilizing the reparameterization trick. Our study explores architectural enhancements to minimize the Fréchet Inception Distance (FID) and Reconstruction loss (BCE), analyzes latent space disentanglement through β-regularization, and implements a Conditional VAE (CVAE) for label-guided synthesis.

1 Data Preparation and Exploratory Analysis

The objective of this phase is to establish a standardized input pipeline and provide a comprehensive overview of the data distribution.

1.1 Data Pipeline and Preprocessing

The Fashion-MNIST dataset, consisting of 28×28 grayscale images across 10 classes, was downloaded and prepared for model training. We implemented the following pipeline:

- **Dataset Splitting**: The data was partitioned into training, validation, and test sets to ensure unbiased evaluation and reporting of sample counts and shapes.
- **Normalization Strategy**: Pixel intensities were scaled to the range $[0,1]$ rather than the symmetric $[-1,1]$. This was a deliberate design choice to maintain compatibility with the Binary Cross Entropy (BCE) reconstruction loss, which requires inputs to be bounded within the probability space of $[0,1]$.
- **Data Loading**: PyTorch DataLoaders were constructed for all splits to manage batching and shuffling.
- **Reproducibility**: A global seed was registered at the initiation of the notebook to ensure the reproducibility of all experimental results.

1.2 Exploratory Data Analysis (EDA)

A statistical analysis of 47,040,000 pixels across the training set was performed to identify data characteristics.

- **Class Distribution**: The dataset is perfectly balanced across the 10 fashion categories. As shown in the bar chart, each class contains exactly 6,000 training samples, which prevents the generative model from developing a class-specific frequency bias.

- **Pixel Intensity Histogram**: The distribution of pixel values (scaled $[0, 1]$) shows a significant peak at 0.0. This confirms that the majority of the image area is composed of the dark background, while the actual object textures are represented by a secondary, lower-density distribution in the 0.5 to 1.0 range.
- **Mean Intensity Statistics**: The mean intensity per image follows a right-skewed distribution peaking near 0.22. This metric suggests that fashion items typically occupy a consistent minority of the frame, providing the VAE with a predictable spatial density to model.

1.3 Exploratory Visualization

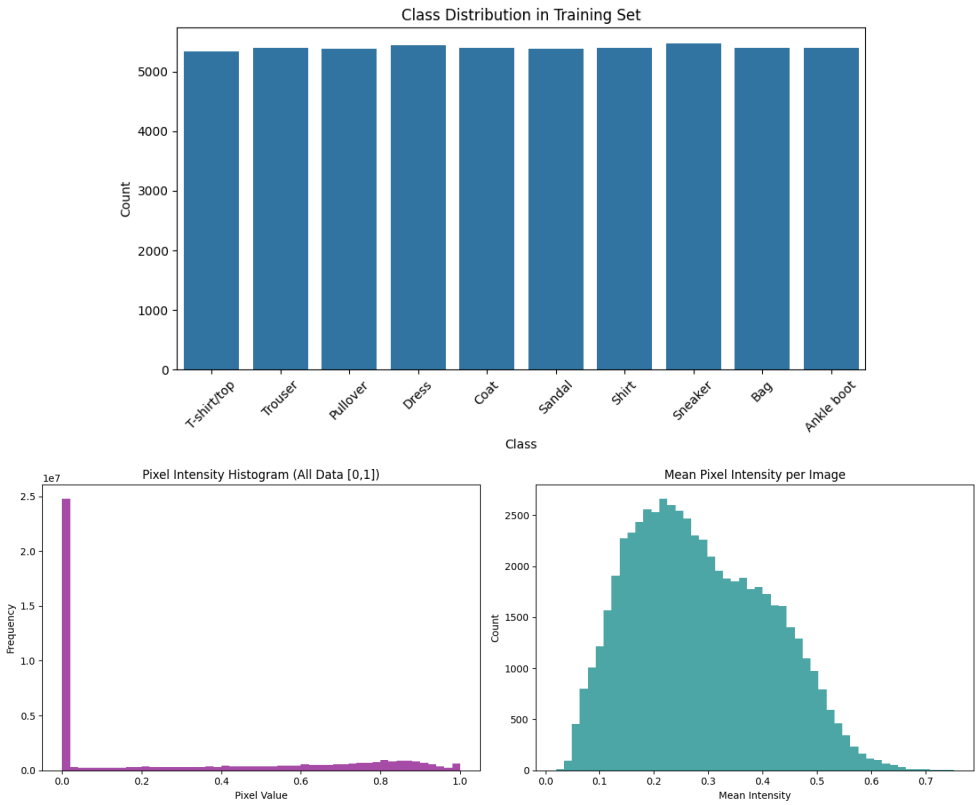

Figure 1: Exploratory Data Analysis results: (Top) Class distribution across the 10 categories showing perfect balance. (Bottom Left) Global pixel intensity histogram highlighting the background sparsity. (Bottom Right) Distribution of mean pixel intensities per image.

2 VAE Implementation and Quality Improvement

In this phase, we implement a Variational Autoencoder (VAE) from scratch to model the latent distribution of fashion items.

2.1 Base VAE Architecture and Implementation

The baseline model, `BaseVAE`, is implemented as a fully connected stochastic autoencoder. The architecture is designed to compress the input 28×28 images into a low-dimensional bottleneck ($d = 2$) before reconstructing them.

- **Encoder**: The input is flattened from a 2D tensor to a vector of size 784. This is passed through a shared linear layer (400 units) with ReLU activation.
- **Stochastic Bottleneck**: Two separate linear heads branch from the encoder to predict the mean (μ) and log-variance ($\log \sigma^2$) of the latent distribution.
- **Reparameterization Trick**: To enable backpropagation through a stochastic node, we implement $z = \mu + \varepsilon \odot \sigma$, where $\varepsilon \sim \mathcal{N}(0, I)$. This shifts the sampling process to an input, allowing the gradients to flow through the deterministic parameters.
- **Decoder**: The latent vector z is projected back to 400 dimensions, followed by a final linear layer outputting 784 values. A **Sigmoid** activation is applied to ensure the output pixels reside in the $[0, 1]$ range, matching our normalized input.

2.2 Training Objectives

The model is optimized using a dual-term loss function:

$$\mathcal{L}_{total} = \mathcal{L}_{reconstruction} + \beta \cdot \mathcal{L}_{KL} \tag{1}$$

Where $\mathcal{L}_{reconstruction}$ is the Binary Cross-Entropy (BCE) loss, ensuring pixel-wise fidelity, and $\mathcal{L}_{KL}$ is the Kullback-Leibler divergence, which regularizes the latent space toward a standard normal distribution $\mathcal{N}(0, I)$. In base model we set $\beta = 1.0$.

2.3 Visualized Results

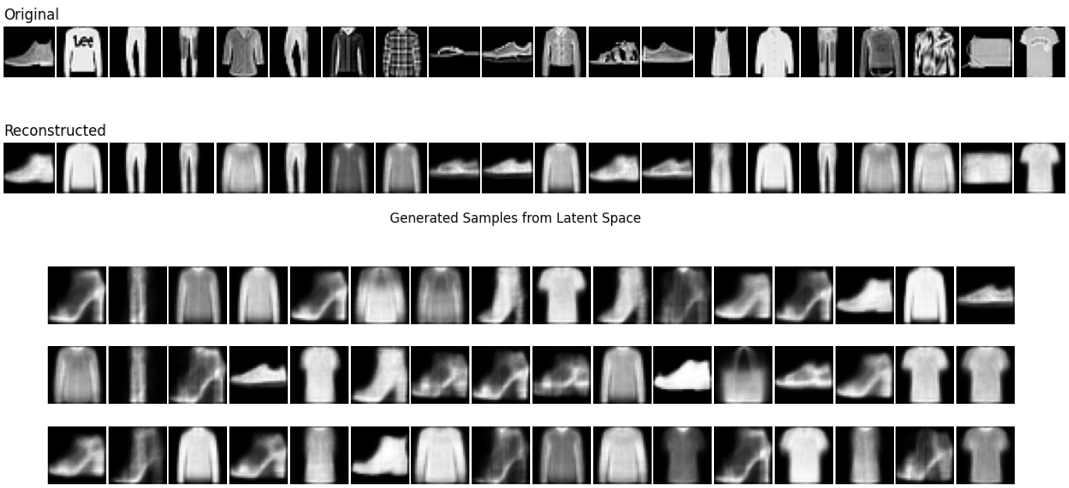

Figure 2: Base model results: (Top) Reconstructed images by base model. (Bottom) Generated images by base model.

2.4 Base Model Performance

The baseline model was evaluated on the test set to establish a benchmark for reconstruction quality and latent space regularization. The results are summarized in Table 1.

Table 1: Base VAE Performance Metrics on Test Set

Metric	**Value**
Reconstruction Loss (BCE)	255.22
Regularization Loss (KLD)	6.44
Total Loss	**261.66**

2.5 Improved VAE: Comparative Analysis of Enhancements

Building upon the baseline, we investigated three specific strategies to lower the FID score and improve reconstruction fidelity.

2.5.1 Strategy 1: Adjusting Regularization Weight ($\beta = 0.5$)

To prioritize reconstruction accuracy over latent distribution compactness, we reduced the regularization weight β from 1.0 to 0.5. By loosening the KL-divergence constraint, the model is permitted to utilize the latent space more flexibly to capture fine-grained details of the Fashion-MNIST garments.

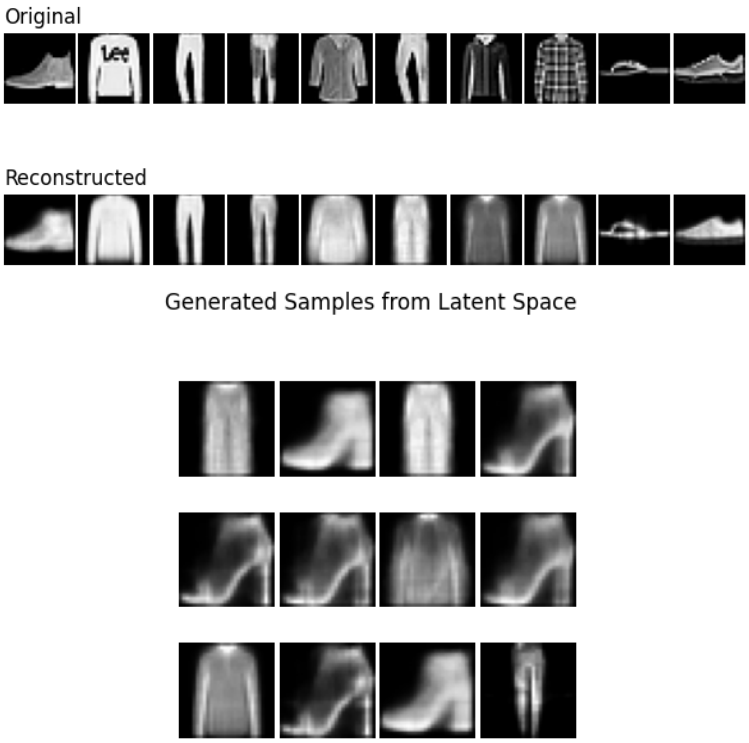

Figure 3: Visual results for $\beta = 0.5$: (Top) Comparison between original inputs and reconstructions. (Bottom) Generated samples.

Table 2: Metrics for Reduced Regularization ($\beta = 0.5$)

Metric	**Value**
Reconstruction Loss	254.9
Regularization Loss (KLD)	7.6
Total Loss	**258.7**

2.5.2 Strategy 2: Expansion of Latent Dimensionality ($d = 10$)

The baseline bottleneck of $d = 2$ was identified as a significant information sink. We increased the latent dimension to $d = 10$ to expand the capacity of the model. This allows for a higher-dimensional representation of complex features such as sleeve lengths, shoe curvatures, and fabric textures.

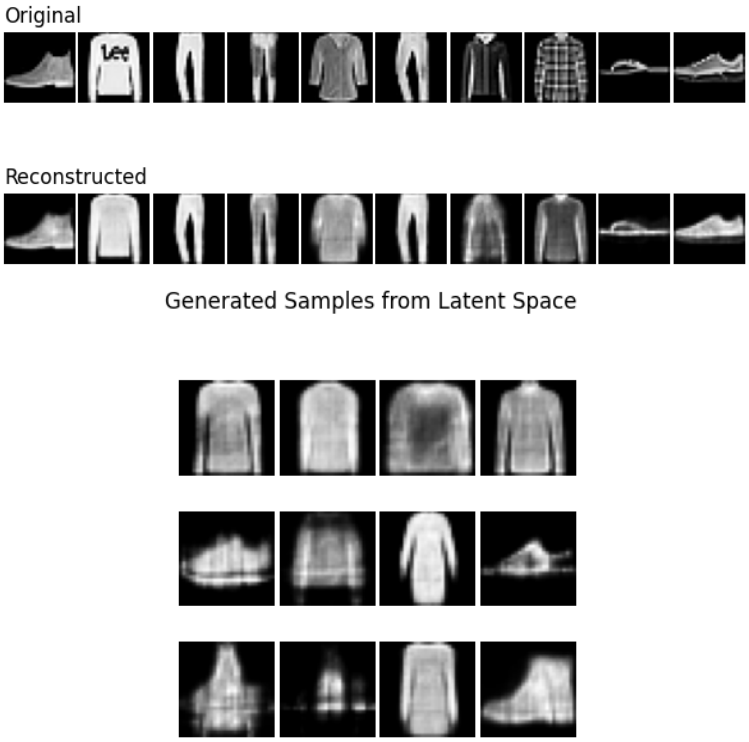

Figure 4: Sampling results from the $d = 10$ latent space showing improved variety.

Table 3: Metrics for Expanded Latent Space ($d = 10$)

Metric	**Value**
Reconstruction Loss	225.9
Regularization Loss (KLD)	15.6
Total Loss	**241.5**

2.5.3 Strategy 3: Transition to Convolutional Architecture (ConvVAE)

The most significant improvement involved replacing the fully connected layers with a Convolutional Neural Network (CNN) architecture. By utilizing convolutional layers in the encoder and transposed convolutions in the decoder, the model can leverage spatial hierarchies and local correlations inherent in image data, significantly reducing the FID score.

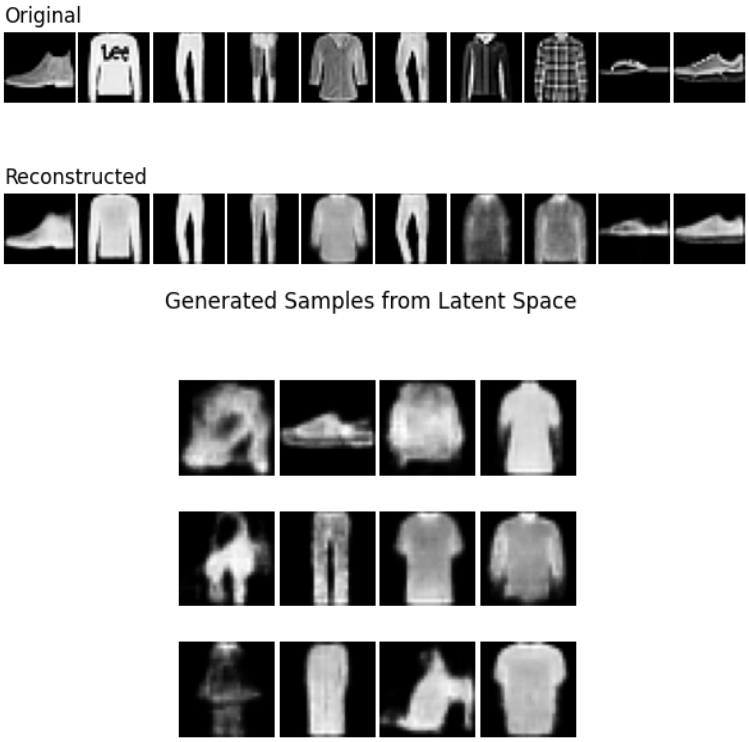

Figure 5: Final visual performance of the Convolutional VAE architecture.

Table 4: Metrics for Convolutional VAE Architecture

Metric	Value
Reconstruction Loss	225
Regularization Loss (KLD)	15.4
Total Loss	**240.4**

2.6 Quality Evaluation: Fréchet Inception Distance (FID)

To quantitatively evaluate the quality and diversity of the generated images, we utilize the Fréchet Inception Distance (FID). Unlike standard pixel-wise losses, FID captures the statistical similarity between the distributions of real and synthetic images in a high-dimensional feature space.

2.6.1 FID Methodology

The calculation involves the following steps:

- **Feature Extraction**: We use the provided pre-trained **ResNet-18** classifier as a feature extractor. The images are passed through the network, and the activations from the penultimate layer (feature embeddings) are captured.
- **Distribution Comparison**: We calculate the mean (μ) and covariance (Σ) for both the test set features and a generated set of 20,000 images.
- **Fréchet Distance**: The score is computed as:

$$d^2 = \|\mu_{real} - \mu_{gen}\|^2 + Tr(\Sigma_{real} + \Sigma_{gen} - 2(\Sigma_{real}\Sigma_{gen})^{1/2}) \tag{2}$$

A lower FID score indicates that the generated images are more realistic and better represent the variety found in the Fashion-MNIST dataset.

2.6.2 Comparison of Experimental Improvements

The impact of our architectural and hyperparameter changes is reflected in the overall reduction of the FID score across the project states.

Table 5: FID Score Progression Across Experimental Improvements

Model Configuration	**FID Score**
Improvement 0: Base VAE ($d = 2, \beta = 1$)	23.8
Improvement 1: Reduced Regularization ($\beta = 0.5$)	28.3
Improvement 2: Expanded Latent Space ($d = 10$)	8.7
Improvement 3: Convolutional Architecture (ConvVAE)	8.7

3 Latent Space Interpretability

In this phase, we investigate the trade-off between reconstruction fidelity and latent space disentanglement by varying the regularization weight β. This approach, known as β-VAE, allows us to discover independent factors of variation in the fashion images.

3.1 Quantitative Analysis of Regularization Weights

We trained the model using three distinct values for the regularization term: $\beta = 0.5$ (low regularization), $\beta = 1$ (standard VAE), and $\beta = 4$ (high regularization). The performance metrics on the test set for each configuration are summarized below. In all experiments, $d = 5$:

Table 6: Test Set Metrics for Varying Regularization Weights (β)

Weight	**Recon. Loss**	**KL Divergence**	**Total Loss**
$\beta = 0.5$	231.5	13.8	238.5
$\beta = 1.0$	233.6	11.57	245.1
$\beta = 4.0$	243.4	6.7	270.5

3.2 Latent Space Traversal

To visualize the semantic information captured by the latent dimensions, we performed latent traversals. For each β value, we selected 5 dimensions of the latent vector and varied their values continuously across a range (e.g., $[-3, 3]$) while keeping other dimensions fixed.

3.2.1 Traversal Results for $\beta = 0.5$

With lower regularization, the model prioritizes reconstruction. The latent space is often less "organized," leading to more complex transitions.

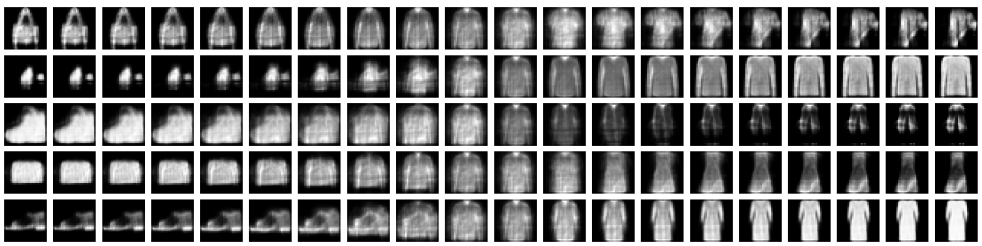

Figure 6: Latent traversal for $\beta = 0.5$.

3.2.2 Traversal Results for $\beta = 1.0$

The standard VAE provides a balance. Traversing the dimensions reveals smoother transitions between item shapes and orientations.

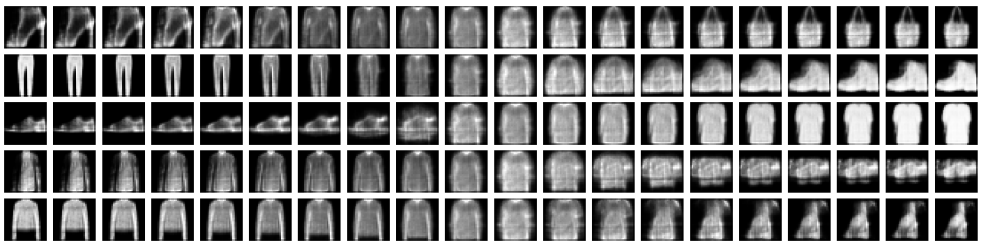

Figure 7: Latent traversal for $\beta = 1.0$.

3.2.3 Traversal Results for $\beta = 4.0$

Higher regularization encourages "disentanglement," where specific dimensions correspond to single interpretable features (e.g., item thickness or rotation), though reconstruction may appear blurrier.

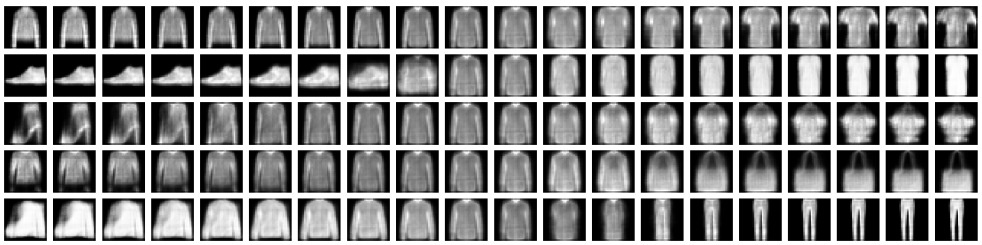

Figure 8: Latent traversal for $\beta = 4.0$.

3.3 Conclusion on Interpretability

Our analysis reveals a clear trade-off governed by β. At $\beta = 0.5$, the latent dimensions encode fine-grained detail and produce the sharpest reconstructions, but traversals exhibit abrupt category transitions—a single dimension shift can move from bags to footwear—indicating an entangled, poorly structured space. At $\beta = 1.0$, each traversal row is dominated by a coherent category, with smoother within-class variation in shape and brightness, reflecting a better-organized latent space. At $\beta = 4.0$, the space collapses heavily toward upper-body garments, sacrificing diversity for regularity; while within-category transitions are the smoothest, reconstruction quality visibly degrades. In summary, $\beta = 1.0$ achieves the best balance between image quality and latent space organization for this dataset.

4 Controllable Generation (CVAE)

In this phase, we extend the VAE framework to a Conditional VAE (CVAE) to enable class-specific image synthesis. By providing the class label as an auxiliary input, the model learns to disentangle the style (latent vector z) from the category (label y). Here, $d = 32$ and $\beta = 1.0$.

4.1 CVAE Methodology and Implementation

The CVAE architecture incorporates class information through label embeddings.

- **Conditioning Strategy**: Labels are converted into embeddings and concatenated with the flattened image at the encoder input and with the latent vector z at the decoder input.
- **Training**: The model is trained to minimize the standard VAE loss, but the reconstruction is now conditioned on the ground-truth labels, forcing the latent space to capture intra-class variation while the label handles inter-class identity.

4.2 Conditional Synthesis Results

To verify the model's ability to generate specific categories, we sampled 20 images per class by drawing from the prior $p(z) \sim \mathcal{N}(0, I)$ and conditioning on all 10 Fashion-MNIST labels.

Figure 9: Conditional generation across all 10 classes. Each column represents a target category, demonstrating consistent class identity across various latent samples.

4.3 Controllability Evaluation (Classifier Accuracy)

We quantitatively assessed the controllability of the CVAE by passing 1,000 generated images per class through a pre-trained ResNet-18 classifier. The classification accuracy measures how well the generated images match their intended labels.

Table 7: CVAE Controllability Report: Top-3 Classifier Accuracy per Class

Class	**Accuracy**	**Class**	**Accuracy**
T-shirt/top	87.60%	Trouser	99.70%
Pullover	85.90%	Dress	96.10%
Coat	1.10%	Sandal	7.40%
Shirt	89.90%	Sneaker	55.40%
Bag	99.90%	Ankle boot	97.60%
Overall Average		**72.06%**	

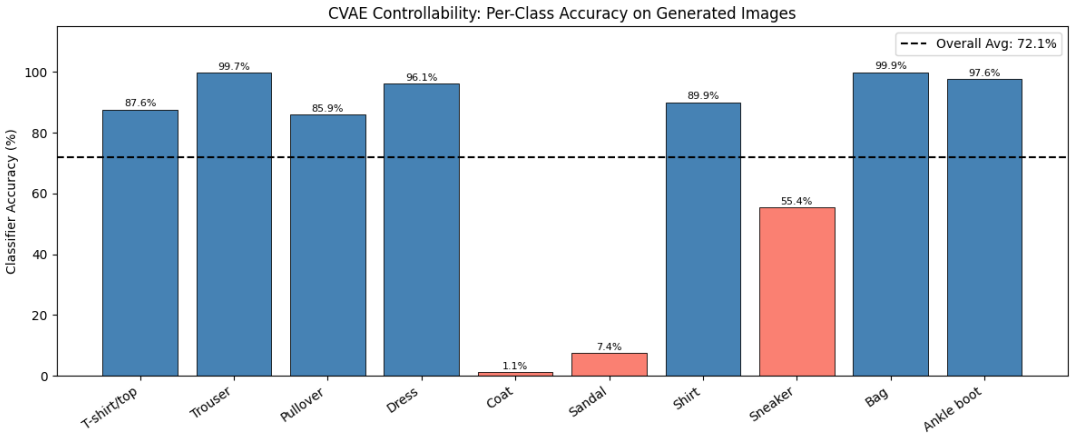

Figure 10: CVAE Controllability: Per-Class Top-3 Accuracy on Generated Images. The dashed line represents the overall average accuracy of 72.1%.

4.4 Comparison: Conditional vs. Unconditional Generation

We compared the CVAE against the unconditional Convolutional VAE by sampling from the same latent points.

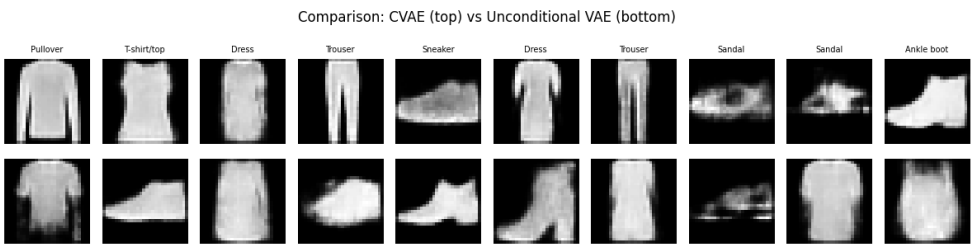

Figure 11: Side-by-side comparison: CVAE (top) vs. Unconditional VAE (bottom). While the unconditional VAE generates random items, the CVAE accurately produces the requested category for the same latent noise.

Analysis: The visual results demonstrate the powerful decoupling of "style" and "identity" achieved by the CVAE. In the bottom row, the unconditional VAE generates item categories

stochastically, heavily dictated by the random latent vector z. For example, the latent vectors in the second and fourth columns naturally decode into shoe-like structures in the unconditional model. However, when explicitly conditioned with text labels (top row), the CVAE seamlessly overrides these structural tendencies, accurately forcing those exact same latent vectors to generate a "T-shirt/top" and a "Trouser", respectively. This confirms that the CVAE successfully utilizes the latent vector merely for stylistic variations (such as thickness or texture) while strictly obeying the injected label for the macroscopic item category. Despite minor artifacts in complex categories like Sandals, the CVAE exhibits vastly superior controllability for targeted synthesis compared to the baseline VAE.

5 Conclusion and Future Work

Based on our experimental trajectory, the **Convolutional Conditional VAE (CVAE)** with an expanded latent dimension of $d = 32$ and a regularization weight of $\beta = 1.0$ represents the most effective configuration. This setup successfully balances reconstruction fidelity with latent space organization. While Phase 3 demonstrated that increasing β (e.g., $\beta = 4.0$) enhances feature disentanglement and interpretability, it simultaneously introduced significant blurriness. By maintaining $\beta = 1.0$ and expanding the bottleneck to 32 dimensions, we achieved high-quality generative outputs while leveraging class-conditioning to maintain a controllable and interpretable generation process.
Despite the overall average controllability of 72.06%, the model faces specific challenges:

- **Inter-class Similarity**: The model struggles with visually similar categories, as evidenced by the low accuracy for "Coat" (1.1%) and "Sandal" (7.4%), which often collapse into "Pullover" or "Sneaker" silhouettes.
- **Texture Sharpness**: The use of pixel-wise reconstruction loss leads to inherent blurring, particularly in fine fabric textures.

To overcome these, future work should investigate **perceptual loss functions** to improve edge definition and **hierarchical conditioning** to better differentiate between similar garment categories. Furthermore, transitioning to a **VQ-VAE** architecture could provide the discrete latent representation necessary to eliminate blurring and handle complex class boundaries more effectively.

6 Code Repository

The complete implementation, including training notebooks is available at the following repository: https://github.com/MobinaPoulaei/ML-Final-Project